TP02: How to Work with Big Data Files (+5GB)

Big Data Analytics Module
University of El Oued

Master 2 - IA & Data Science

1 Educational Objectives

- Understand the Out-Of-Memory (OOM) limitations of traditional data loading methods when working with large datasets (+5GB).
- Implement Stream Processing and Chunking techniques using standard Pandas workflows.
- Leverage Parallel and Lazy Computing using the Dask Framework.
- Benchmark and profile performance metrics (Execution Time vs. Memory Footprint).

2 Environment & Dataset Setup

- **Dataset:** New York City Taxi Trip Data (`train.csv`)
- **File Size:** 5.31 GB
- **System Total RAM:** 15.82 GB
- **Target Column:** `fare_amount` (calculating `sum` and `mean`)

3 Implementation Summary & Tasks

- **Task 1 (Standard Pandas):** Attempted to load the entire 5.31 GB dataset at once, resulting in heavy resource usage and a very slow execution time (**222.49 s**).
- **Task 2 (Pandas Chunking):** Processed the file sequentially in chunks of 100,000 rows, keeping memory stable ($\sim$4787 MB) with a moderate execution time (**100.75 s**).
- **Task 3 (Dask Framework):** Utilized 89 partitions with *Lazy Evaluation* and *Parallel Computing*, achieving the fastest execution time (**31.1 s**) and optimal memory management.

4 Performance Benchmark Results

Here are the actual benchmark results captured from the execution:

Method	Execution Time (s)	Peak RAM (MB)	Status
Standard Pandas	222.49 s	4,776.4 MB	Loaded (High Cost)
Pandas Chunking	100.75 s	4,787.4 MB	Success (Sequential)
Dask DataFrame	**31.1 s**	**4,370.7 MB**	**Success (Fast & Parallel)**

5 Conclusion

- Traditional Pandas fails to handle large-scale data efficiently due to memory overhead.
- Chunking solves the memory overflow issue but remains limited by sequential execution speed.
- **Dask** is proven to be the optimal solution for Big Data Analytics, combining parallel computing and lazy evaluation for maximum performance.